Dr. David Raj M

Ph.D. (Mathematics) · M.Tech. (CSE, AI & ML)

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Research profiles

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Google Scholar
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Web of Science
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Research interests

Linear algebra & generalized inverses · Graph theory & network science · Fuzzy logic & decision making · Machine learning & deep learning · Image processing & computer vision · Scientific computing & algorithms

Profile

Data scientist and mathematician bridging rigorous linear algebra, graph theory, and fuzzy modelling with machine learning for healthcare and engineering applications. Experienced in publication, faculty development programmes, and student-centred teaching with modern LMS tools.

Technical skills

..groups.map(g => skill-line(g))

Education

2022–2025 **M.Tech. Computer Science & Engineering (AI & ML)**
Vellore Institute of Technology, Chennai
94% (aggregate)

2016–2020 **Ph.D. in Mathematics**
Manipal Academy of Higher Education, Manipal
Thesis: _Generalized Inverses and its Applications_

2014–2015 **M.Phil. in Mathematics**
Bharathidasan University, Trichy — 81%

2012–2014 **M.Sc. in Mathematics**
Bharathidasan University, Trichy — 71.27%

2009–2012 **B.Sc. in Mathematics**
St. Joseph's College (Autonomous), Trichy — 82.50%

Academic appointments & administration

2021–present **Assistant Professor**
Department of Mathematics, School of Advanced Sciences, VIT Chennai
Teaching, research supervision, and interdisciplinary collaboration in AI/ML and applied mathematics.

2025–present **Assistant Director, Ranking and Accreditation**
Vellore Institute of Technology, Chennai

Feb–May 2021 **Assistant Professor**
St. Joseph's College, Bangalore

2019–2021 **Assistant Professor**
Sikkim Manipal Institute of Technology, Sikkim
Rainfall distribution modelling (machine learning & statistics).

2015–2018 **Junior Research Fellow**
Department of Data Science (formerly Statistics), MAHE, Manipal
Generalized inverses, iterative algorithms, SageMath/Python implementations.

Awards & fellowships

2015 **UGC NET-JRF (Mathematical Sciences) — 112/314**

Key publications & research highlights

Selected impact-focused works; full bibliography below and on Google Scholar.

- **Low-light image enhancement using fuzzy methods** — Novel parametrized intuitionistic fuzzy framework for low-light enhancement with bounded parameter search and HSI-space modelling.
  Applied Soft Computing, 2025
- **Optimal graph embedding and wirelength calculation** — Algorithms for minimum linear arrangement and embedding of graph products into grids and related host topologies.
  The Journal of Supercomputing, 2024
- **COVID-19 vaccine decision system** — Energy-based bipolar intuitionistic fuzzy digraph model for multi-criteria vaccine selection in healthcare settings.
  Applied Soft Computing, 2023
- **Intuitionistic fuzzy transportation problem solver** — Branch-and-bound technique for fully intuitionistic fuzzy transportation problems with improved computational efficiency.
  Journal of Intelligent and Fuzzy Systems, 2023
- **ICU mortality prediction using deep learning** — Semi-supervised clustering of MICU cohorts (MIMIC-II) with deep survival models for mortality risk stratification.
  Informatics in Medicine Unlocked, 2023

Peer-reviewed publications & book chapters

Selected list; full bibliography on Google Scholar. Author name emphasized.

2025 **Yagavi R.**, **Mahesh Kumar C.V.**, Arumugam S., **David Raj M.** Stress of a Graph—Theory, Algorithms and Applications. In *Graph Theory and Its Applications: Topological Indices of Graphs I*, pp. 9–24. Selçuk University Press.

2025 Maheshkumar C.V., **David Raj M.**, Saraswathi D. On generalized Sugeno's class generator and parametrized intuitionistic fuzzy approach for enhancing low-light images. *Applied Soft Computing* 172, 112865.

2025 Vincy G. Caroline, **David Raj M.** On the optimal layout of $\left(K_p - C_p\right)^n$ into grid and certain structures. *Scientific Reports* 15, 17114.

2024 Vincy G. Caroline, **David Raj M.** Minimum linear arrangement and embedding of $(K_{11} - C_{11})^n$ into specific graphs with optimal wirelength calculation. *The Journal of Supercomputing* 81, 218.

2023 M.K. Lintu, **David Raj M.**, Asha Kamath. Mortality prediction on unsupervised and semi-supervised clusters of medical intensive care unit patients based on MIMIC-II database. *Informatics in Medicine Unlocked* 39, 101264.

2023 Deva Nithyanandham, Felix Augustin, **David Raj M.**, Nagarajan Deivanayagam Pillai. Energy based bipolar intuitionistic fuzzy digraph decision-making system in selecting COVID-19 vaccines. *Applied Soft Computing*, 110793.

2023 J. Jansi Rani, S. Dhanasekar, **David Raj M.**, A. Manivannan. On solving fully intuitionistic fuzzy transportation problem via branch and bound technique. *Journal of Intelligent & Fuzzy Systems* 44(4), 6219–6229.

2023 M.K. Lintu, Amitha Puranik, Kalesh M. Karun, **David Raj M.**, Asha Kamath. Non linear relationship between BMI and Sepsis among ICU patients: analysis of the MIMIC-III real world database. *Journal of Clinical and Diagnostic Research*.

2022 **David Raj M.**, M.K. Lintu, S.N.S. Acharya, Asha Kamath. Mortality prediction of patients admitted to surgical intensive care unit using deep learning based survival models. *Malaysian Journal of Science* 41(3), 44–48.

2018 K. Manjunatha Prasad, **David Raj M.** Bordering method to compute core-EP inverse. *Special Matrices* 6, 193–200.

2018 K. Manjunatha Prasad, **David Raj M.**, M. Vinay. An iterative method to find Core-EP inverse. *Bull. Kerala Math. Assoc.* 16(1), 139–152.

2017 Ravindra B. Bapat, S.K. Jain, K. Manjunatha Prasad, **David Raj M.** Outer inverses: characterization and applications. *Linear Algebra and its Applications* 528, 171–184.

Conference proceedings & edited volumes

2025 Maheshkumar C.V., **David Raj M.**, Saraswathi D. Parametrized interval valued intuitionistic fuzzy set for low light image enhancement. *ISCMI 2025*, pp. 239–243.

2024 Yogesh Chettri, **David Raj M.**, M.F. Abdul Rahim. Propagation of two-soliton in a compressible hyperelastic rod. *ICNDA 2024*, Springer, pp. 295–307.

2023 **David Raj M.**, A. Felix, Vimaleshwaran S. Prediction of rainfall using machine learning in Chennai region. *Contemporary Research Trends in Mathematics*, pp. 90–106.

2022 Barsha Pradhan, Nikhil Pal, **David Raj M.** Bifurcation of nucleus-acoustic superperiodic and supersolitary waves in a quantum plasma. *Nonlinear Dynamics and Applications*, Springer, pp. 43–53.

2021 S. Kiran, **David Raj M.** Determination of best-fit probability distribution of rainfall data in Sikkim, India. *Water Security and Sustainability*, Springer, pp. 65–77.

Manuscripts under review

2025 Maheshkumar C.V. et al. Adaptive low light image enhancement using bipolar fuzzy set. *IEEE Transactions on Fuzzy Systems* (communicated).

2025 Maheshkumar C.V., **David Raj M.**, Saraswathi D. Exploring angle and amplitude encoding in quanvolutional neural network for MNIST classification. *Philosophical Magazine* (communicated).

2025 Maheshkumar C.V., **David Raj M.**, Saraswathi D. Intuitionistic fuzzy-based enhancement of low-light color images using image fusion and grayscale processing. *Computers and Electrical Engineering* (communicated).

2025 Yagavi R., **David Raj M.** Stress centrality in weighted and fuzzy networks. *Applied Soft Computing* (communicated).

Invited talks & resource person

2023	**Guest lecture — Survival Analysis. VIT Chennai.**
2023	**Five-day STTP — The Role of Linear Algebra in ML. SVCE, India.**
2023	**Workshop for CBSE teachers — Computation in school mathematics. Bala Vidya Mandir, Chennai.**
2022	**FDP — Advanced LaTeX (class & style files). VIT Chennai.**
2022	**Hands-on Python for mathematicians. Periyar University, Salem.**
2022	**Special talk — Image processing with Python. Sanghamam College, Tamil Nadu.**
2021	**FDP — Mathematics of Machine Learning. ANJAC, Sivakasi.**
2018	**Workshop — SageMath: matrices, graphs and networks. CARAMS, MAHE Manipal.**

Events organised

2023	**FDP on Linear Algebra and its Applications. VIT Chennai.**
2022	**One-day online workshop — Typing question papers in LaTeX. VIT Chennai.**
2022	**FDP on Recent Advances in Mathematics. VIT Chennai.**
2019	**National workshop — Typing scientific documents in LaTeX. SMIT, Sikkim.**
2020	**National workshop on special matrices and stochastic games. SMIT, Sikkim.**

Conference presentations

2023	**A note on the computation of core-EP inverse. MAHE, Manipal.**
2019	**Best-fit probability distribution for rainfall in North Sikkim. ICWSS, SMIT.**
2017	**Computational methods to find core-EP inverse. ICLAA, MAHE.**
2016	**Iterative method to find core-EP inverse. Manipal Student Research Colloquium.**
2016	**Bordering method to compute core-EP inverse. NCLA, SMU Sikkim.**
2015	**Characterization of outer inverses and absorption law. ICMPS, Goa.**

Teaching experience

2021–present	**Applied Multivariate Data Analysis; Python for Data Science; Linear Algebra; Complex Analysis; Calculus (VIT Chennai).**
2021	**Undergraduate mathematics (St. Joseph's College, Bangalore).**
2019–2021	**Engineering Mathematics II–IV; Discrete Mathematics; Numerical Methods; Computational Lab (SMIT).**
2015–2018	**Linear Algebra; Real Analysis; Measure Theory (MAHE, as JRF teaching support).**

Delivery: classroom (chalk & digital), LMS (Canvas), recorded lectures, and open educational content (YouTube: DavidRajM).

Research supervision

Supervising doctoral and postgraduate scholars in graph embedding, centrality measures, fuzzy modelling, and image processing; guiding undergraduate project teams in ML and data science. *Update scholar names and counts in `content.typ` for formal submissions.*

Languages

English	Fluent — read, write, speak
Tamil	Native
Hindi	Conversational
Kannada	Conversational

Recent achievements

Administrative leadership, process automation, industry consultancy, and training engagements (2025–2026).

Ranking, accreditation and quality assurance

Assistant Director, Ranking and Accreditation · VIT Chennai (2025–present)

- Lead cross-departmental data collection, validation, and submission for national and international rankings (NIRF, QS, Shanghai, and allied frameworks).
- Support faculty target-setting; analyse attainment against institutional goals and deliver improvement feedback.
- Conduct faculty workshops on outcome-based education—CO formulation, PO mapping, and attainment analysis.
- **Monthly target review automation:** unified departmental dashboards and growth-vs-target presentations; reduced manual compilation from 2 days to 2 hours.
- **Quality Review Report automation:** weekly activity compilation for valedictory release (including morning sessions); reduced preparation from 1 day to 1 hour.
- **NBA data quality validator:** programmatic checks on 500+ faculty profiles before submission, surfacing invalid records without manual review.
- **CO compliance tracker:** defaulter lists for pending course-outcome calculations by subject, generated in minutes.

Academic process automation

Department of Mathematics · VIT Chennai

- Full-stack **timetable scheduler** for slot-based timetabling (regular and flexi slots).
- Full-stack **project review management system**—coordinators assign reviewers, capture scores, and receive automated reports.
- **Result analytics** for internal examinations: graphical CO attainment analysis and slow-learner identification.
- Workflow utilities including automated extraction of slot-wise marks from consolidated mark files.

Industry consultancy

Kadant India Private Limited · June 2025

- Delivered software to measure nano-tooth outlines from images, overlay digital dimensions on source imagery, and produce layout-ready digital models. Techniques: preprocessing, edge detection, intersection geometry, deep learning, and web interface.

Training and professional development

Corporate and academic engagements

- Regular delivery in Python, R, web stacks, automation, LaTeX/Typst, data structures, ML/DL, generative AI, and LLMs.
- Feb 2026 — Faculty development programme, Dayananda Sagar University: deep learning, generative AI, and transformer models.
- Feb 2026 — MSc Data Science (VIT Chennai): agentic AI and autonomous intelligent systems.
- Mar 2026 — Faculty training (VIT Chennai): data structures and deep learning.

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